

Practice Problem Set 1 (38)

1. In the space below, create a friendship graph (point and line) with the provided friendship ties (8 points). Who do you think is the most popular? Who is the least popular? How many nodes and edges are there?(You can just use the first initial, they all have different initials to start their names)

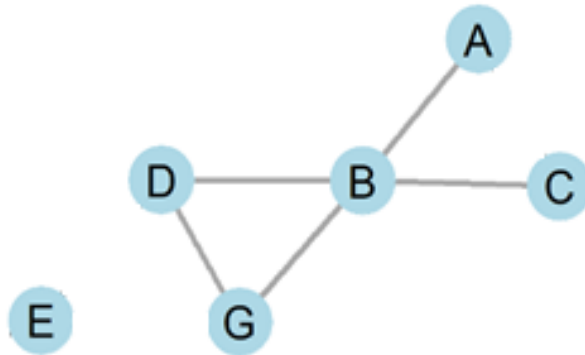
Alex and Patrick are friends.
Xin and Wanda are friends.
Tanya and Wanda are friends.
Xin and Myles are friends.
Myles and Ricardo are friends.
Ricardo and Xin are friends.
Sasha and Ricardo are friends.

Tanya and Youssef are friends.
Denzel and Youssef are friends.
Denzel and Sasha are friends.
Sasha and Tanya are friends
Myles and Sasha are friends.
Sasha and Patrick are friends.
Myles and Patrick are friends.

Additional Questions

Most Popular? _____ (1 point for trying)
Least Popular? _____ (1 point for trying)
How many nodes in the Graph? _____ (1 pt)
How many edges in the Graph? _____ (1 pt)
Name two nodes that are adjacent _____ (1 pt)
Name two nodes that are non- adjacent _____ (1 pt)

2. Describe the set that reflects the following point and line representation of graph **G**? (6 points for the set)



Additional Questions

How many nodes are in the graph? _____ (1 pt)

How many edges are in the graph? _____ (1 pt)

Name two nodes that are adjacent _____ (1 pt)

Name two nodes that are non- adjacent _____ (1 pt)

What is a path sequence starting with origin node D and ending with destination node C? (1 pt)

3. Answer the following questions about dyads and triads. (1 pt each, 6 total)

- a. A and B are co-workers. What type of dyad is this?
- b. H and J don't know each other. What type of dyad is this?
- c. L, M, and N are all friends with one another. What type of triad is this?
- d. O and P are friends, but neither are friends with Q. What type of triad is this?
- e. C, D, and E are not friends with one another. What type of triad is this?
- f. X knows Y, and X knows Z, but Z and Y don't know each other. What type of triad is this?

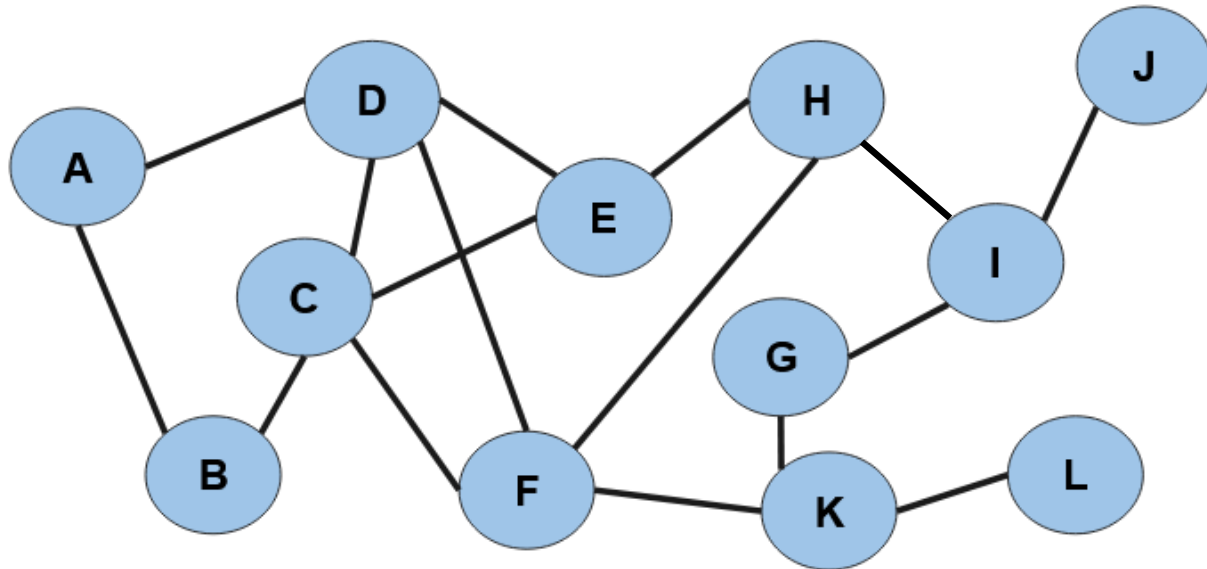
5. Think about a Network (3 points total)

This class is about networks, so first, pick a topic that you are personally interested in. Can you take your idea and start to study it as a network? (Don't overthink this, it's cool, one-word answers are fine. Like I'm interested in supply routes, airports are entities, and flights are the connectons). Use the opportunity to explore something you are genuinely interested in!

- a. What is the topic you are interested in?
- b. What are some actors/entities that you could study relating to this topic?
- c. What are the relationships/connections that connect these actors/entities?

4. Practice Problems: The Ministry of Silly Walks (8pts)

Both J and L would like to get approval for funding the development of their new idea, which can only be approved by person A. The bureaucracy to A though is a little convoluted, not terribly well organized, and the ties are non-directed.



- a) What is the shortest path sequence for person **L** to get their idea passed up the chain of command for approval? What is the path length (2pt)
- b) What is the shortest path sequence for person **J** to get their idea passed up the chain of command for approval? What is the path length? (2pt)
- c) What is the other term for the shortest path between two nodes? (1pt)
- d) What is an example of a walk of length 5? (1pt)
- e) List a cycle starting with node F. (1pt)
- f) List a trail starting with Node F, then going next to Node C, coming back to C at some point, and ending with Node A. (1pt)